

Abhishek Bisht

✉ abhishekbisht1429@gmail.com 📞 +91 8979588935 🆔 0000-0002-9953-7531 🌐 abhishek-bisht1429 📧 abhishekbisht1429

EDUCATION

Master of Science in Computer Science and Engineering by Research

Jan 22 - May 24

International Institute of Information Technology, Hyderabad, India

CGPA - 9.0/10

- **Thesis Title:** Design and Implementation of Efficient Personal Health-Records Sharing in IoMT using Searchable Symmetric Encryption, Blockchain and IPFS
- **Advisor:** Prof. Ashok Kumar Das
- **Research Lab:** Center for Theory Security & Algorithmic Research (CSTAR)
- **Courses:** Distributed Systems, Principles of Information Security, Research in Information Security

Bachelor of Technology in Computer Science & Engineering

Aug 17 - May 21

University of Petroleum and Energy Studies, Dehradun, India

CGPA- 8.77/10

- **Courses:** Design and Analysis of Algorithms, Programming and Data Structures, Operating Systems, Computer Architecture, Computer Networks, Software engineering

RESEARCH EXPERIENCE (H-Index: 4)

Research Interests

Machine Learning, Artificial Intelligence, Cyberphysical Systems, Cyber Security, AI/ML Security, Blockchain

Published Works

1. Prakash Tekchandani, Abhishek Bisht, Ashok Kumar Das, Neeraj Kumar, Marimuthu Karuppiah, Pandi Vijayakumar, Youngho Park **"Blockchain-Enabled Secure Collaborative Model Learning using Differential Privacy for IoT-Based Big Data Analytics"** *IEEE Transactions on Big Data (IF 7.5)* 2024
DOI: 10.1109/TBDATA.2024.3394700
2. Basudeb Bera, Abhishek Bisht, Ashok Kumar Das, Bharat Bhargava, David KY Yau, Pascal Lorenz, Biplab Sikdar **"BioKA-ASVN: Biometric-Based Key Agreement Scheme for Air Smart Vehicular Networks Using Blockchain Service"** *IEEE Transactions on Vehicular Technology (IF 6.8)* 2024
DOI: 10.1109/TVT.2024.3380392
3. Sandip Roy, Abhishek Bisht, Ashok Kumar Das, Sachin Shetty, M Shamim Hossain, Prakash Tekchandani, Neeraj Kumar, Marimuthu Karuppiah, Pandi Vijayakumar, Youngho Park **"Age of correlated information-optimal dynamic policy scheduling for sustainable Green IoT devices: A multi-agent deep reinforcement learning approach"** *Elsevier - Internet of Things (IF 6.0)* 2024
DOI: 10.1016/j.iot.2024.101141
4. Abhishek Bisht, Ashok Kumar Das, Debasis Giri **"Personal health record storage and sharing using searchable encryption and blockchain: A comprehensive survey"** *Wiley - Security and Privacy (IF 1.5)* 2024
DOI: 10.1002/spy2.351
5. Abhishek Bisht, Ashok Kumar Das, Dusit Niyato, Youngho Park **"Efficient Personal-Health-Records Sharing in Internet of Medical Things Using Searchable Symmetric Encryption, Blockchain and IPFS"** *IEEE Open Journal of the Communications Society (IF 6.3)* 2023
DOI: 10.1109/OJCOMS.2023.3316922
6. Sudeep Ghosh, SK Hafizul Islam, Abhishek Bisht, Ashok Kumar Das **"Provably secure public key encryption with keyword search for data outsourcing in cloud environments"** *Elsevier - Journal of Systems Architecture (IF 3.7)* 2023
DOI: 10.1016/j.sysarc.2023.102876
7. JoonYoung Lee, MyeongHyun Kim, KiSung Park, SungKee Noh, Abhishek Bisht, Ashok Kumar Das, Youngho Park **"Blockchain-based data access control and key agreement system in IoT environment"** *MDPI - Sensors (IF 3.4)* 2023
DOI: 10.3390/s23115173
8. Roushan Kumar, Rupendra Kumar Pachauri, Pankaj Badoni, Deepak Bharadwaj, Udayveer Mittal, Abhishek Bisht **"Investigation on parallel hybrid electric bicycle along with issuer management system for mountainous region"** *Elsevier - Journal of Cleaner Production (IF 9.2)* 2022
DOI: 10.1016/j.jclepro.2022.132430
9. Diksha Mehta, Janhvi Joshi, Abhishek Bisht, Pankaj Badoni **"Efficiency Analysis of Pre-trained CNN Models as Feature Extractors for Video Emotion Recognition"** *Intelligent Systems and Sustainable Computing* 2022
DOI: <https://zenodo.org/records/5515542>

Under Review

1. *Prithwi Bagchi, Abhishek Bisht, Basudeb Bera, Ashok Kumar Das, Bharat K. Bhargava, Biplab Sikdar* ”**Quantum Secure Lattice-Based Aggregate Signature for IoT-Enabled Healthcare using Blockchain**” *IEEE Transactions on Dependable and Secure Computing (IF 7.3)* **2024**
2. *Basudeb Bera, Abhishek Bisht, Ashok Kumar Das, Sandip Roy, Sachin Shetty, Biplab Sikdar* ”**Adversarial Attacks Resilient Machine Learning-Enabled Access Control Scheme for Consumer Electronics in IoT-Based Applications**” *IEEE Transactions on Consumer Electronics (IF 4.3)* **2024**

TEACHING EXPERIENCE

- Teaching Assistant** **Jan 23 - Apr 23**
International Institute of Information Technology, Hyderabad, India
• Worked as a TA for the course Principles of Information Security with 100+ students in Spring 2023 semester.
- IIIT/Conduira Instructor** **Nov 22 - Dec 22**
International Institute of Information Technology, Hyderabad, India
• Worked as an instructor for *Research Teaser* initiated by IIIT for undergraduate research in collaboration with Conduira.
- Head Teaching Assistant** **Aug 22 - Dec 22**
International Institute of Information Technology, Hyderabad, India
• Worked as Head TA for the course Software Systems Development with 100+ students in Fall 2022.

PROFESSIONAL EXPERIENCE

- Software Development Engineer II** **Mar 2024 - Present**
AI Group, AMD, Hyderabad, India
• Part of core team for development of ONNX VitisAI Execution Provider.
• Actively worked on code-base of state-of-the-art Ryzen-AI Strix NPU to support various AI models.
- Co-op/Intern** **Jul 2023 - Jan 2024**
AI Group, AMD, Hyderabad, India
• Worked on TCF-DAP which allows debugging of programs targeting AMD adaptive computing platforms such as Alveo.
• Implemented Test-Infrastructure for TCF-DAP as a part of Vitis IDE.

TECHNICAL SKILLS

Programming/Scripting Languages:

- *Standard Languages:* C/C++, Java, Python, JavaScript
- *Scripting Languages:* Shell (Bash), PowerShell
- *Domain Specific Languages (DSLs):* make, CMake, Groovy

System Programming:

- Compiler (LLVM), LLVM-MLIR, ONNX Runtime, TCF-DAP Protocol, Socket Programming

Application Development:

- *Server Backend:* Django, Spring-boot, Micro-service Architecture, REST Architecture
- *Android App Development:* MVVM Architecture, Android Multi-Threading Framework, Gradle
- *Tools:* Git/Github, CLion, IntelliJ IDEA, PyCharm

ML/AI:

- *Concepts:* Neural Networks, Convolutional Neural Networks (CNN), Deep Q-Networks (DQN), Multi-agent-Reinforce Learning (MARL)
- *Libraries/Frameworks:* NumPy, Pandas, Scikit-learn, Keras, Tensorflow, PyTorch, Jupyter, Matplotlib, ONNX

CERTIFICATIONS/ACHIEVEMENTS

- **MITx 6.86x, EdX** - Machine Learning with Python - From Linear Models to Deep Learning.
- **Coursera Neural Networks and Deep Learning** - Completed a course on Neural Networks by Andrew NG.
- **GATE 2021** - All India Rank 852 in Graduate Aptitude Test (GATE) for CSE in 2021.
- **Technical Head** - Managed both the App team and the Web team of UPES ACM Student Chapter.
- **Android App Development Head** - Managed the App Development team of UPES ACM Student Chapter.